Individual Assingment 5 JM

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2026-05-10

Set Up

Packages

```
# geberal and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
```

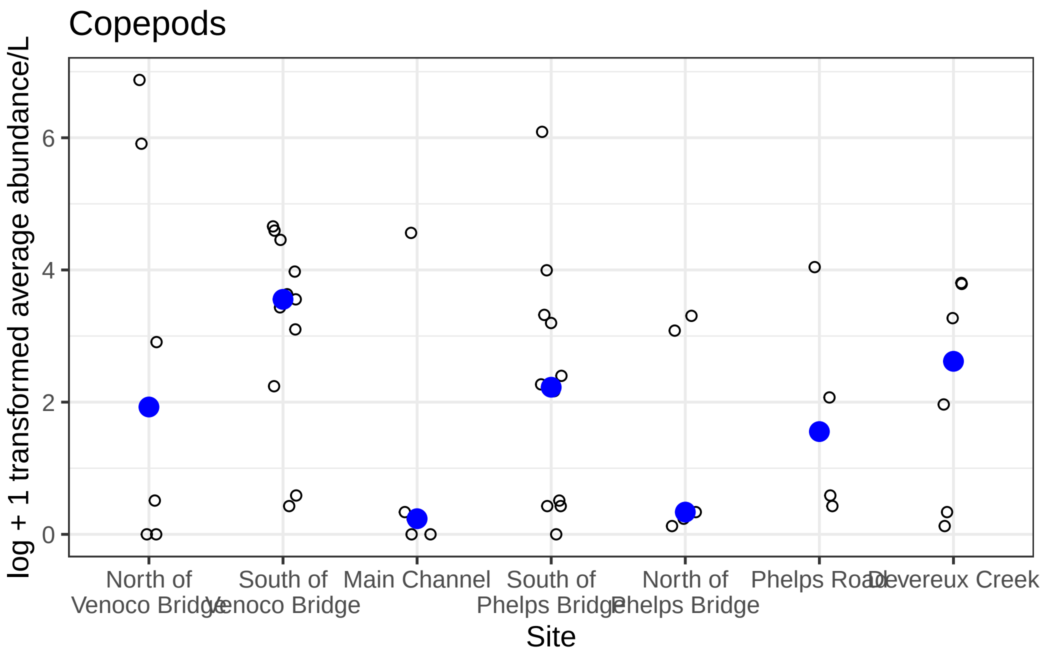
```
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
```

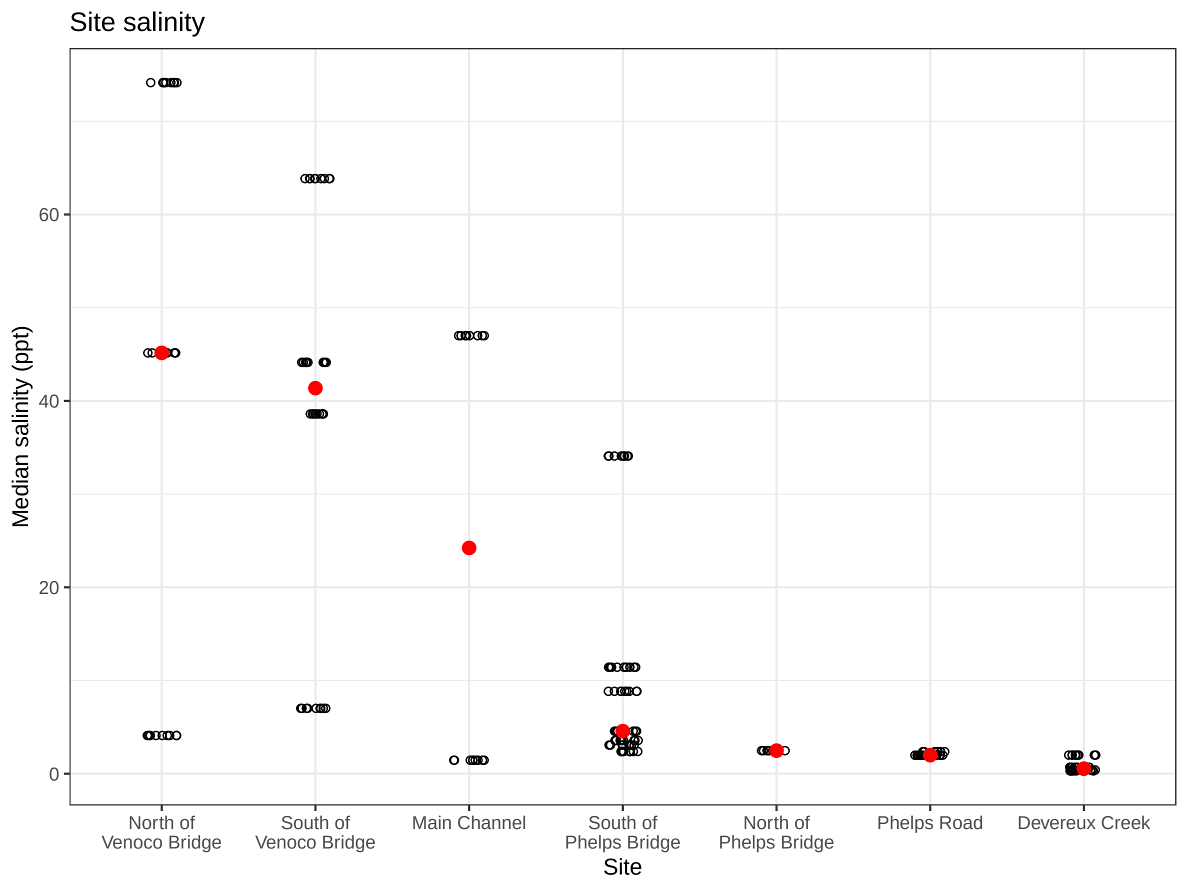
```
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```
copepods_plot <- ggplot(data = copepods,
                   mapping = aes(x = site_full, y = log_ave_lit)) +
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  stat_summary(geom = "point", fun = "median", size = 3, color = "blue") +
  scale_x_discrete(labels = label_wrap(15)) +
  labs(x = "Site", y = "log + 1 transformed average abundance/L", title = "Copepods") +
  theme_bw()

copepods_plot
```




```
salinity <- ggplot(data = aq_ins_clean,
       mapping = aes(x = site_full,
                     y = med_sal)) +
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  scale_x_discrete(labels = label_wrap(width = 15)) +
  stat_summary(fun = median,
               color = "red",
               size = 0.5) +
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  theme_bw()

salinity
```



Problems

1. Visualizing differences across sites in major taxon abundance

a. Planning figure

-X axis: site (site_full) -Y axis: ostracod abundance (log + 1 transform) -geometry to represent average abundance/L: geom_jitter() -geometry to represent medians: stat_summary()

b. Wrangle data

```
ostracods <- aq_ins_clean |>
  filter(taxon == "ostracod") |>
```

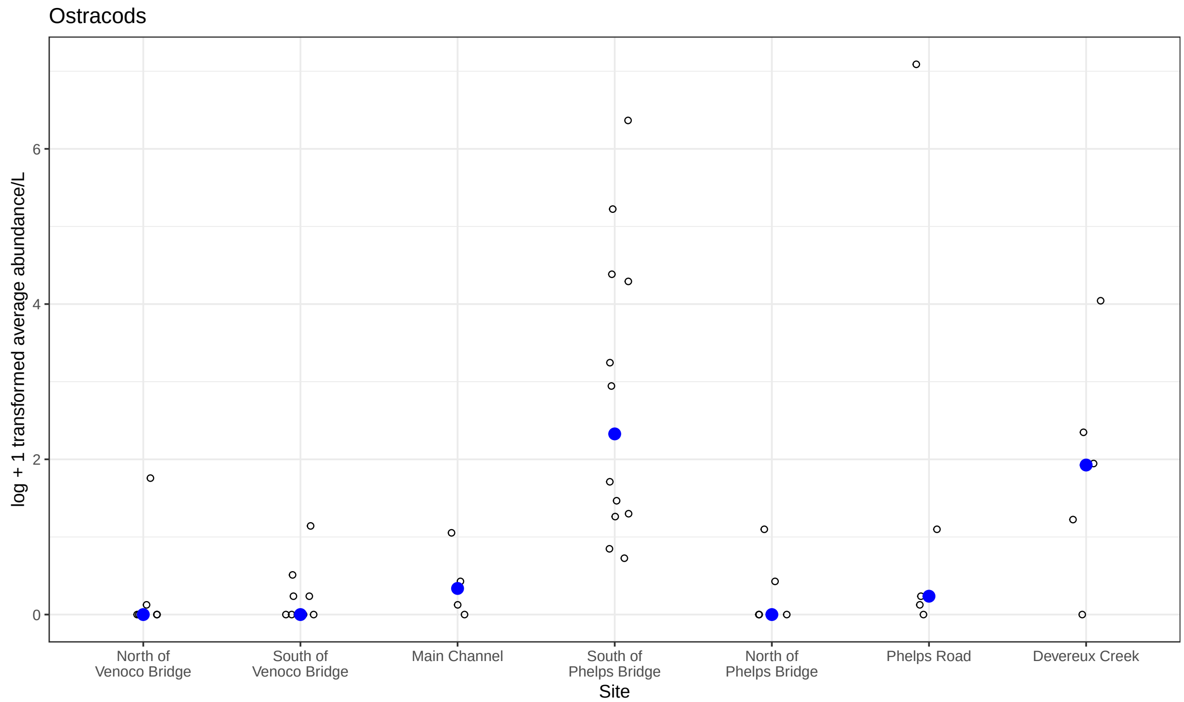
```r
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-05-16_NVBR 2023-05-16 00:00:00 NVBR  ostr~   0.133   9.29  NA       9.83
 2 2022-05-11_NDC  2022-05-11 00:00:00 NDC   ostr~   0       9.11   0.398   9.4
 3 2023-02-28_NPB  2023-02-28 00:00:00 NPB   ostr~   2.53   NA     NA      NA
 4 2021-11-12_NPB  2021-11-12 00:00:00 NPB   ostr~  79.2    NA     NA      NA
 5 2022-08-08_NPB2 2022-08-08 00:00:00 NPB2  ostr~   2       7.68   2.36    2.43
 6 2022-01-21_NEC  2022-01-21 00:00:00 NEC   ostr~   0      NA     NA      NA
 7 2023-06-06_NPB1 2023-06-06 00:00:00 NPB1  ostr~   0.533   7.48  NA       2.38
 8 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71   1.98    9.59
 9 2022-08-15_NMC  2022-08-15 00:00:00 NMC   ostr~   0.133   7.61  NA      NA
10 2023-11-14_NPB  2023-11-14 00:00:00 NPB   ostr~  18      NA     NA      NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Coding figure

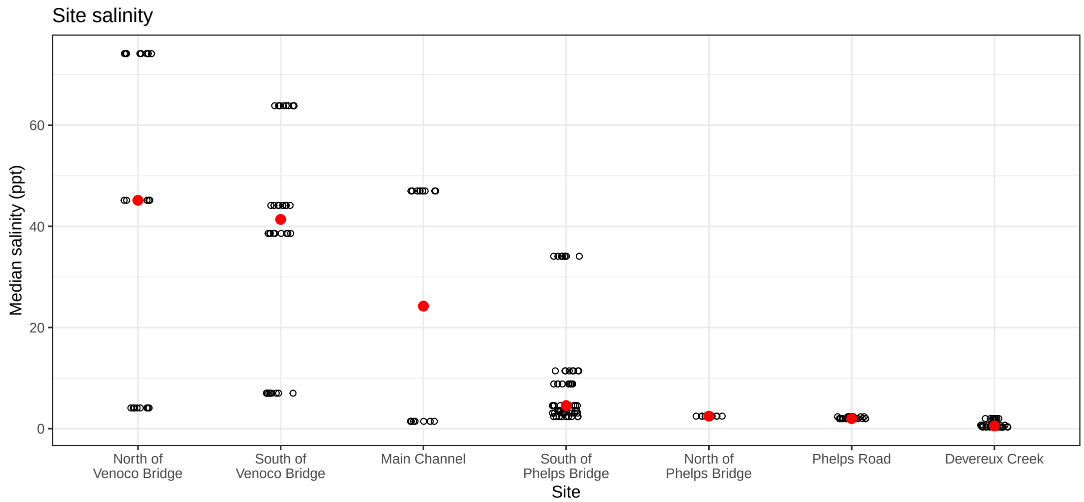
```r
ostracod_plot <- ggplot(data = ostracods,
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # small open circles for individual measurements
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # blue filled point for median
  stat_summary(geom = "point",
               fun = "median",
               size = 3,
               color = "blue") +
  # wrap long site names
  scale_x_discrete(labels = label_wrap(15)) +
  # labels
  labs(x = "Site",
```

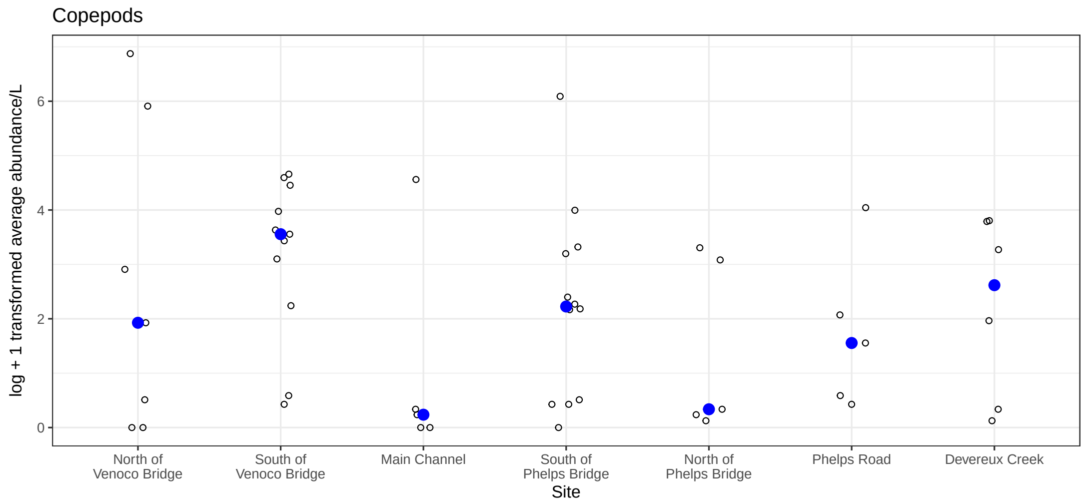
```
      y = "log + 1 transformed average abundance/L",
      title = "Ostracods") +
  # same theme as key
  theme_bw()

ostracod_plot
```

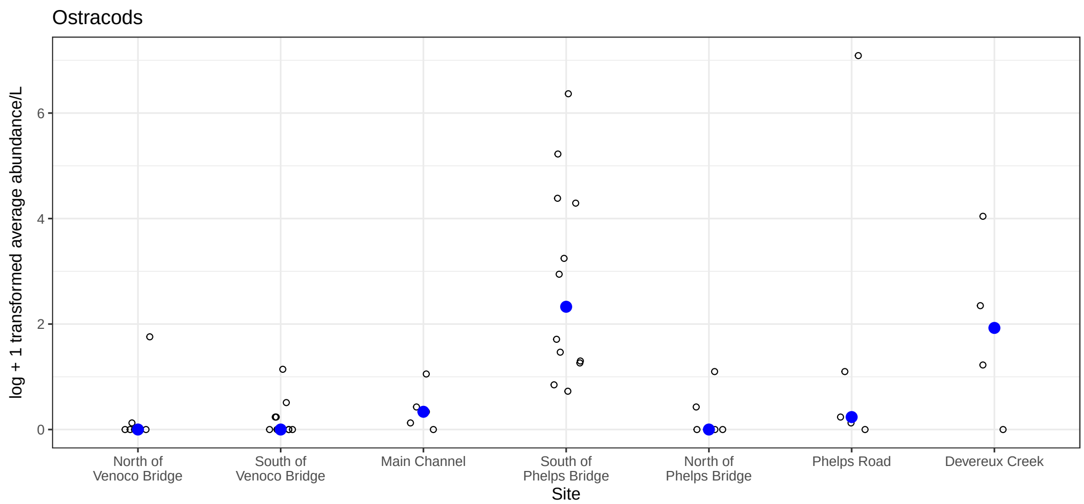


d. Adding ostracod panel

```
salinity / copepods_plot / ostracod_plot
```


Site salinity
Median salinity (ppt)
0
20
40
60
North of
Venoco Bridge
South of
Venoco Bridge
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site


Copepods
log + 1 transformed average abundance/L
0
2
4
6
North of
Venoco Bridge
South of
Venoco Bridge
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site


Ostracods
log + 1 transformed average abundance/L
0
2
4
6
North of
Venoco Bridge
South of
Venoco Bridge
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site

e. Interpretation of the figure

The abundance of copepods north and south of Venoco Bridge and at Phelps Road is higher than the abundance of ostracods, the abundance of each is similar in the Main Channel, south of Phelps Bridge, north of Phelps Bridge, and at Devereux creek. There does not seem to be a strong correlation between of the copepods and ostracods with the salinity.

2.

a.

- X axis: salinity (med_sal)
- Y-axis: log transformed average abundance/L (log_ave_lit)
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~taxon, scales = "free"), aq_ins_clean

b. Wrangle data

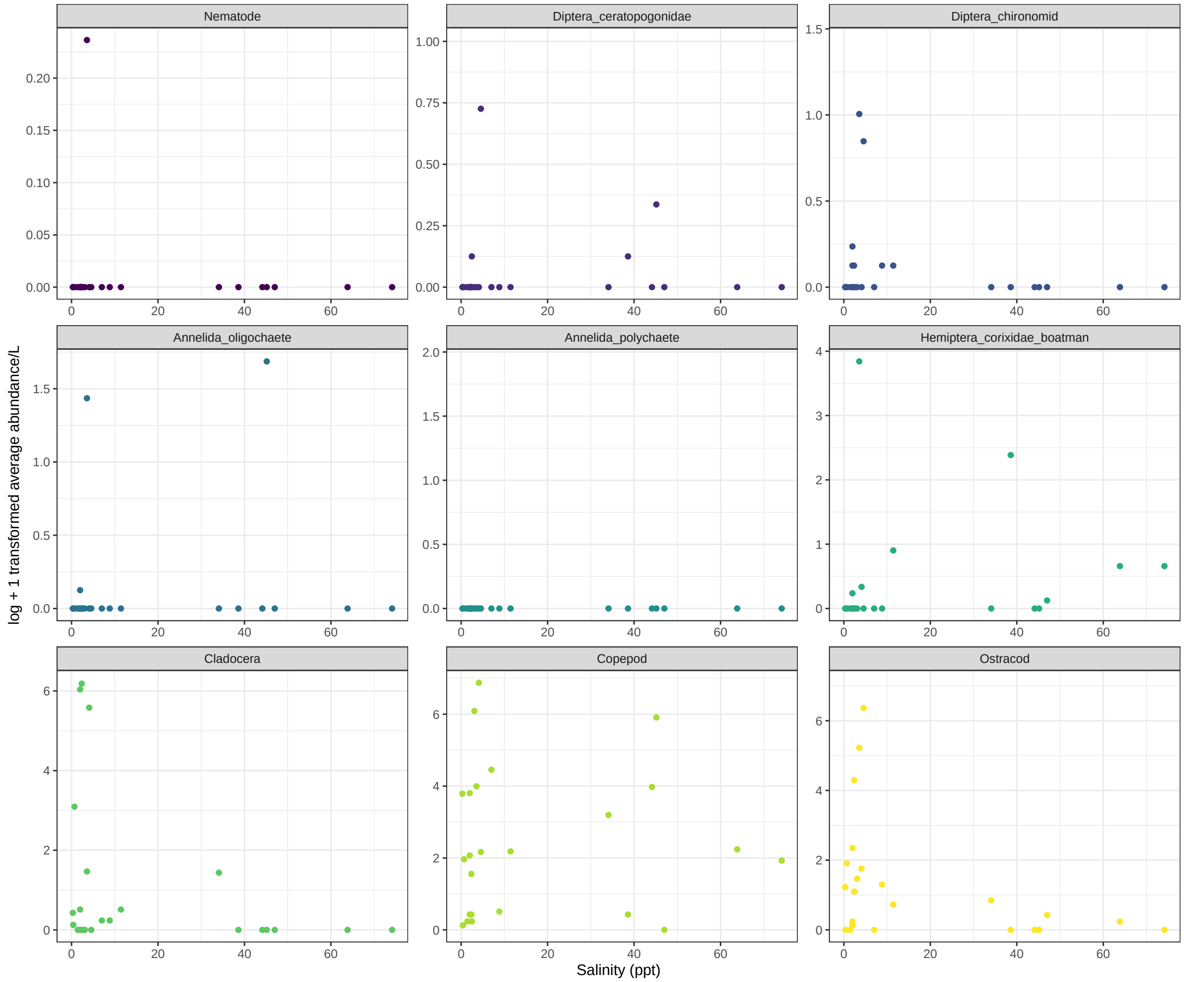
```r
abundance_salinity <- aq_ins_clean |>
  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon to title case
  mutate(taxon = str_to_title(taxon)) |>
  # order taxon levels by maximum abundance
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = "max"))

# display 10 random rows
slice_sample(abundance_salinity, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <fct>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  Cope~   0.533   7.71   1.98    9.59
 2 2023-05-04_NEC  2023-05-04 00:00:00 NEC   Dipt~   0       8.05  NA       8.55
 3 2022-08-08_NPB2 2022-08-08 00:00:00 NPB2  Nema~   0       7.68   2.36    2.43
 4 2022-02-21_NPB  2022-02-21 00:00:00 NPB   Clad~   3.2     7.39  34.1     0.34
 5 2023-05-16_NVBR 2023-05-16 00:00:00 NVBR  Cope~  17.3     9.29  NA       9.83
 6 2022-05-05_NEC  2022-05-05 00:00:00 NEC   Dipt~   0       9.45  63.8     7.3
 7 2022-05-11_NDC  2022-05-11 00:00:00 NDC   Nema~   0       9.11   0.398   9.4
```

```
 8 2023-05-04_NMC  2023-05-04 00:00:00 NMC   Ostr~   1.87    8.32  NA      10.3
 9 2022-05-12_NPB2 2022-05-12 00:00:00 NPB2  Anne~   0      NA     NA      NA
10 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1  Hemi~   0      NA     NA      NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c.

```
abundance_salinity_plot <- ggplot(data = abundance_salinity,
                                  mapping = aes(x = med_sal,
                                                y = log_ave_lit,
                                                color = taxon)) +
  # points for each observation
  geom_point() +
  # separate panel for each taxon with free scales
  facet_wrap(~taxon, scales = "free") +
  # readable axis labels
  labs(x = "Salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # different theme
  theme_bw() +
  # different colors
  scale_color_viridis_d() +
  # remove legend
  theme(legend.position = "none")

abundance_salinity_plot
```



d. Interpreting figure

Nematodes spike about a salinity of 5ppt but stay around 0 for every other salinity value; a similar pattern is seen in the annelida oligochaete where is spikes at arond 5ppt and 45ppt but is around 0 for every other value. Diptera ceratopogonidae has spikes around 5-10ppt and 30-50ppt. Cladocera is much more abundant at lower salinities ranging from 1-10ppt. The copepod graoh seems a bit more random/scattered but has higher abundance from 1-10ppt anf around 40ppt.